

Data Session 11: Did the Machinery Work?

84-355 Democracy's Data | The Election Administration and Voting Survey

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Everything here runs as given. Your job is to read what comes out and say what it means.

Where did that number come from?

“Thousands of mail ballots were rejected.”

Every session so far has used the **output** of an election — who won, who voted, who gave money. None of them asked whether the machinery worked.

Somebody does ask. Every two years the **Election Assistance Commission** sends a survey to every election office in the country and asks: how many ballots did you mail out, how many came back, how many did you reject and why, how many provisional ballots did you issue, how many precincts did you run, how many poll workers did you find.

It is the closest thing America has to a service record for its own elections, and it is free. Today we open it.

Part 1 — What one row is

```
nat <- read.csv("data/national.csv", stringsAsFactors = FALSE)
st  <- read.csv("data/states.csv",   stringsAsFactors = FALSE)
u   <- read.csv("data/units.csv",   stringsAsFactors = FALSE)

c(jurisdictions = sum(st$jurisdictions), states_and_territories = nrow(st))
```

```
#>      jurisdictions states_and_territories
#>                6461                    56
```

6,461 rows. One per election jurisdiction. And that word is the first problem.

```
rbind(head(u[, c("state", "jurisdictions", "voters", "voters_per_jurisdiction")], 4),
      tail(u[, c("state", "jurisdictions", "voters", "voters_per_jurisdiction")], 4))
```

```
#>      state jurisdictions    voters voters_per_jurisdiction
#> 1      VT              247    361604                1464
#> 2      ME              497    842447                1695
#> 3      WI             1851   3434185                1855
#> 4      NH              320    829090                2591
#> 53     CA              58  16164330               278695
#> 54     DC              1     328871               328871
#> 55     AK              1     340981               340981
#> 56     PR              1   1283628               1283628
```

Wisconsin reports 1,851 jurisdictions. Texas reports 254. Wisconsin does not have seven times as much election administration as Texas — it runs elections through **townships**, and Texas runs them through **counties**. New England is the same: Vermont’s typical jurisdiction has about 1,500 voters; California’s has about 279,000.

So a sentence like “the average jurisdiction rejected 90 ballots” is close to meaningless, because the average is taken over units that are not the same kind of thing. **Anything you compute per jurisdiction, across states, is wrong.** Totals are fine. Rates are fine. Averages over rows are not.

Part 2 — The national picture

```
nat[, c("item", "total", "reported_by", "pct_reporting")]
```

```
#>              item      total reported_by pct_reporting
#> 1      Total voters 158211780      6449      99.8
#> 2      Total precincts    177708      6451      99.8
#> 3 Mail ballots transmitted to voters 66867671      6445      99.8
#> 4      Mail ballots returned by voters 47957093      6131      94.9
#> 5 Mail ballots returned undeliverable    932533      5843      90.4
#> 6      Mail ballots rejected    584463      6361      98.5
#> 7      Provisional ballots cast    1736209      5654      87.5
#> 8 Provisional ballots counted in full    1199465      5621      87.0
#> 9      Provisional ballots rejected    436258      5162      79.9
```

158 million voters, 177,708 precincts, 66.9 million mail ballots sent out.

Now look at the last two columns, which are the ones nobody quotes. **Not every jurisdiction answers every question.** Total voters is reported by 99.8% of them; provisional ballots rejected, by **79.9%**.

That means the 436,258 figure is a sum over four-fifths of the country, and the missing fifth is not random — small offices with less staff are likelier to skip items. **Every national total on this page is a floor, not a count.**

```
# Missing is coded -99 (not reported) and -88 (not applicable), not blank.
# Summing the raw column would quietly mix real counts with these codes.
nat[nat$variable %in% c("E1a", "E1d"), c("item", "total", "pct_reporting")]
```

```
#>              item      total pct_reporting
#> 7      Provisional ballots cast 1736209      87.5
#> 9 Provisional ballots rejected 436258      79.9
```

Part 3 — Rejected mail ballots

```
sent <- nat$total[nat$variable == "C1a"]
ret  <- nat$total[nat$variable == "C1b"]
rej  <- nat$total[nat$variable == "C9a"]

c(transmitted = sent, returned = ret, never_returned = sent - ret,
   rejected = rej, pct_of_returned = round(100 * rej / ret, 2))
```

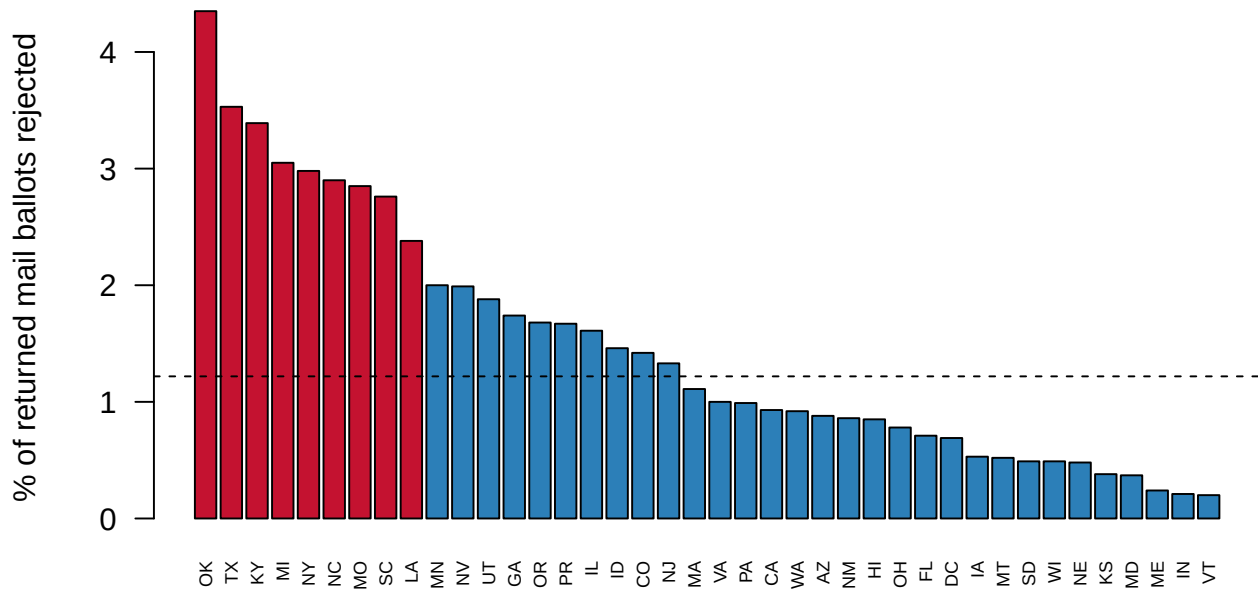
```
#>      transmitted      returned never_returned      rejected pct_of_returned
#> 66867671.00    47957093.00    18910578.00    584463.00          1.22
```

584,463 mail ballots were rejected — 1.22% of those returned. That sounds small, and nationally it is. Now do it by state.

```
st$rate <- round(100 * st$mail_rejected / st$mail_returned, 2)
big <- st[st$mail_returned > 100000, ]
rbind(head(big[order(-big$rate), c("state", "mail_returned", "mail_rejected", "rate")], 5),
       head(big[order( big$rate), c("state", "mail_returned", "mail_rejected", "rate")], 4))
```

```
#>      state mail_returned mail_rejected rate
#> 40      OK      103025      4477 4.35
#> 48      TX      398270      14049 3.53
#> 20      KY      120400       4076 3.39
#> 25      MI      2081265     63561 3.05
#> 38      NY      862737     25750 2.98
#> 52      VT      237565       467 0.20
#> 18      IN      1607247     3432 0.21
#> 24      ME      215753       511 0.24
#> 23      MD      747040      2796 0.37
```

```
b <- big[order(-big$rate), ]
barplot(b$rate, names.arg = b$state, las = 2, cex.names = 0.55,
        col = ifelse(b$rate > 2, "#C41230", "#2c7fb8"),
        ylab = "% of returned mail ballots rejected")
abline(h = 100 * rej / ret, lty = 2)
```



Oklahoma rejects 4.35% of returned mail ballots. Vermont rejects 0.20%. Among states handling more than 100,000 mail ballots, that is a **twenty-two-fold difference** in the chance your ballot is thrown out.

That is not voters behaving differently in Oklahoma. It is **rules** — witness requirements, signature matching, postmark deadlines, and whether the state lets you fix a problem after the fact. The same act, performed identically, is counted in one state and not in another.

Now the part of that ranking nobody checks. Every rate in it is *rejected* ÷ *returned*: two separate survey items, each with its own response rate, neither answered by the whole country.

Before you run the next chunk, answer yes or no. A missing denominator sounds as though it should behave the way missing data usually does — scatter the rates about a bit, some too high, some too low, no particular direction. **Does it?** Write down yes or no, and if no, write down which way it pushes.

```
nat[nat$variable %in% c("C9a", "C1b"),
     c("item", "total", "reported_by", "pct_reporting")]
```

```
#>               item      total reported_by pct_reporting
#> 4 Mail ballots returned by voters 47957093      6131      94.9
#> 6           Mail ballots rejected  584463      6361      98.5
```

```
# States reporting mail ballots sent, but zero returned
hole <- st[st$mail_sent > 0 & st$mail_returned == 0,
          c("state", "voters", "mail_sent", "mail_returned", "mail_rejected")]
hole
```

```
#>   state  voters mail_sent mail_returned mail_rejected
#> 2    AL 2272911   140558           0         1779
#> 8    CT 1820891   129996           0         2642
#> 28   MP   12610    1168           0           13
#> 29   MS 1225176   28558           0           0
```

No. It pushes one way, and then it takes states out of the table altogether.

The denominator is measured worse than the numerator. The rejection count comes from 98.5% of jurisdictions, the returned count from only 94.9%. Wherever an office reports the one and not the other, the denominator is short and the rate comes out **too high** — not noisily, always upward.

At state level it stops being subtle. Some 4 states and territories, with 5,331,588 registered voters between them, report sending mail ballots while reporting that **zero** came back — and 3 of them go on to report rejecting some of the none that returned. Alabama sent 140,558 and rejected 1,779.

You cannot compute a rate from that, so Alabama is not at the top of the ranking above and not at the bottom. **It is not in the ranking.** A zero and a missing value look identical once they are in a spreadsheet, and only arithmetic that cannot work reveals the difference. “Oklahoma is 22 times worse than Vermont” is a statement about the 40 states that answered both questions. **Non-response is not noise around the estimate; it decides who is eligible to be compared at all** — and the offices likeliest to skip an item are the small, understaffed ones.

Part 4 — Why they were rejected, and who says

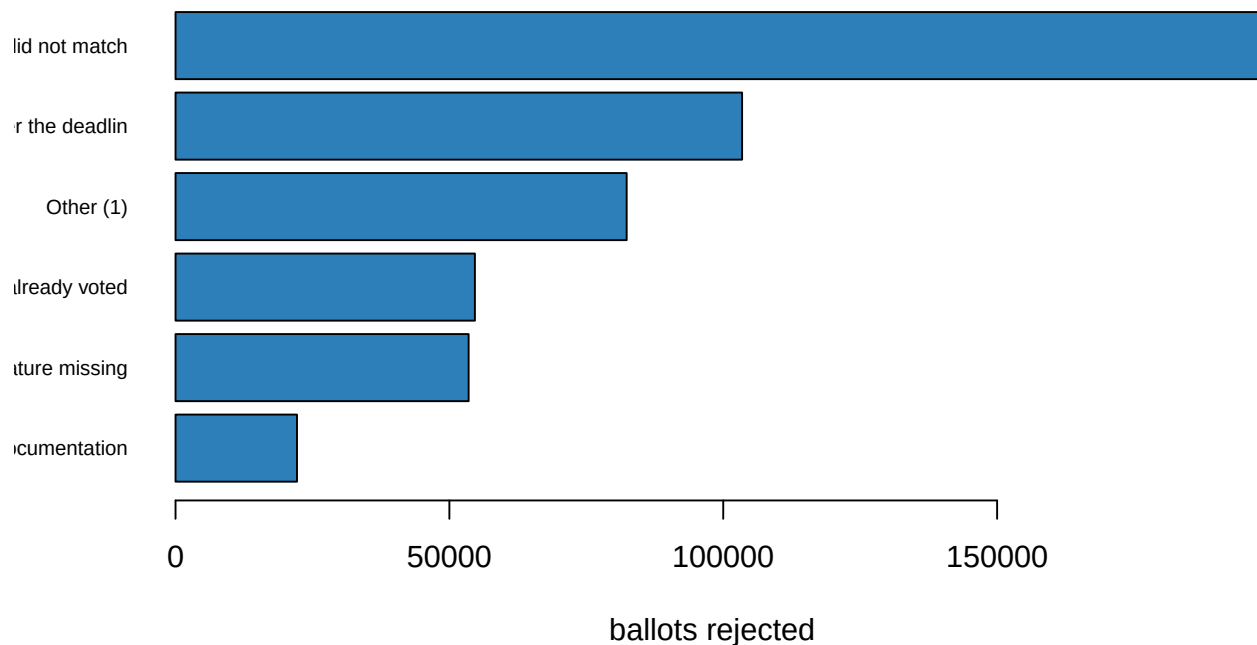
```
r <- read.csv("data/reasons.csv", stringsAsFactors = FALSE)
head(r[, c("reason", "ballots", "pct_of_rejected", "reported_by")], 8)
```

#>		reason	ballots	pct_of_rejected	reported_by
#> 1		Signature did not match	197979	33.9	3222
#> 2		Received after the deadline	103441	17.7	6093
#> 3		Other (1)	82366	14.1	4452
#> 4		Voter already voted	54659	9.4	5284
#> 5		Voter signature missing	53511	9.2	4198
#> 6		Missing documentation	22175	3.8	3284
#> 7		Witness signature missing	17623	3.0	2794
#> 8		Voter not eligible	12018	2.1	5026

The single largest reason is a signature that did not match: 197,979 ballots, 34% of every mail ballot rejected in the country. A human being compared your handwriting to a specimen on file and decided it was not you.

Second is arriving late — 103,441. Third, at 82,366, is “Other.”

```
top <- head(r, 6)
barplot(rev(top$ballots), horiz = TRUE, names.arg = rev(substr(top$reason, 1, 26)),
        las = 1, cex.names = 0.7, col = "#2c7fb8", xlab = "ballots rejected")
```



Now the column that matters most, and it is not the count:

```
r[, c("reason", "ballots", "reported_by")][c(1:3, 17:19), ]
```

```
#>               reason ballots reported_by
#> 1      Signature did not match 197979      3222
#> 2      Received after the deadline 103441      6093
#> 3              Other (1) 82366      4452
#> 17 Multiple ballots in one envelope      668      4208
#> 18 No resident address on envelope      635      1959
#> 19 No ballot application on file      131      2140
```

The total rejected is reported by 6,361 jurisdictions. The reasons are reported by as few as 845. You can find out *how many* ballots were thrown out far more reliably than *why* — which is exactly backwards from what anyone trying to fix the problem would need.

```
c(stated_total = rej, sum_of_reasons = sum(r$ballots),
  difference = rej - sum(r$ballots))
```

```
#>   stated_total sum_of_reasons   difference
#>         584463         581498           2965
```

The reasons do very nearly add up — a gap of about 3,000 out of 584,463. **That is a good sign and it does not rescue the point above.** The categories account for the total; the *jurisdictions* reporting each category do not cover the country.

Part 5 — Provisional ballots

```
cast <- nat$total[nat$variable == "E1a"]
full <- nat$total[nat$variable == "E1b"]
prej <- nat$total[nat$variable == "E1d"]
c(cast = cast, counted_in_full = full, rejected = prej,
  pct_rejected = round(100 * prej / cast, 1))
```

```
#>           cast counted_in_full      rejected    pct_rejected
#>      1736209.0      1199465.0      436258.0           25.1
```

A provisional ballot is what you get when you turn up and something is wrong — you are not on the list, you lack the right ID, you are at the wrong place. It is the system's safety net.

One in four is rejected. 436,258 people.

That is a far higher failure rate than mail, and it is much less discussed, because provisional ballots are issued to exactly the people least likely to have anybody advocating for them. Note also that this figure is reported by only **79.9%** of jurisdictions, so the real number is larger.

Part 6 — Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Provisional by state. Compute $100 * \text{prov_rejected} / \text{prov_cast}$ for states with more than 10,000 provisional ballots. The spread is wider than mail. Which states stand out, and what would you need to know before blaming either the voters or the rules?

Option B — Mail take-up. Part 3 shows 66.9 million ballots sent and 48.0 million returned — 19 million never came back. Compute that share by state ($1 - \text{mail_returned}/\text{mail_sent}$). Where is it highest, and is an unreturned mail ballot evidence of anything?

Option C — The reporting gap. Rank the reasons by **reported_by** instead of by **ballots**. Which reasons are best reported and which worst? Is there a pattern in what election offices find easy to say?

Option D — Your own question. 6,461 jurisdictions, 535 columns in the source file, and a codebook that tells you what every one of them means.

What this lab cannot tell you

- **Whether a rejection was correct.** The file records that a ballot was rejected and the reason recorded by the office. It does not record whether the signature really was different, or whether the voter was told, or whether they could fix it.
- **Who the rejected voters were.** No demographics. Research linking rejection to race and age exists and uses voter files — Session 6's data — matched to administrative records, not this survey.
- **A clean national total.** Response is 80–99% depending on the item, and the non-response is not random. Everything here is a floor.
- **Comparisons across states without care.** Different rules, different units, different definitions of the same word. That is not a defect in the survey; it is a fact about American elections that the survey is honest enough to expose.
- **What happened in 2026.** This is 2024. The 2026 EAVS will not be published until mid-2027 — long after the election we watched in Session 10.

On the discussion board

By **Friday, 11:59 p.m.** A sentence or two is enough.

Three that would make good posts:

- Your ballot is 22 times likelier to be rejected in Oklahoma than in Vermont. Is that a problem, and whose?
 - One in four provisional ballots is rejected. Should the safety net work better than that, or is that what a safety net looks like?
 - The count of rejections is well reported; the reasons are not. Who benefits from that asymmetry, if anyone?
-

Where the data came from

U.S. Election Assistance Commission, **Election Administration and Voting Survey, 2024, Version 2** (released 12 February 2026), 6,461 jurisdictions and 535 columns, with the official codebook. <https://www.eac.gov/research-and-data/datasets-codebooks-and-surveys>

Version 2, not Version 1 — V1 was corrected, and the EAC publishes an errata note listing what changed. A lab built on V1 would disagree with the EAC’s own report.

Three traps handled in `data/build-data.R`: missing values are coded **—99 and —88** rather than left blank, so anything summing a raw column gets nonsense; the file contains **embedded newlines** inside comment fields, so counting lines gives 17,013 for a file with 6,461 rows; and **“jurisdiction” means different things in different states.**